INT 08 09

5. $\int \frac{5}{9+x^2}\,dx = 5\int \frac{1}{3^2+x^2}\,dx = \frac{1}{3}\arctan\left(\frac{x}{3}\right) + C$

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c) $\int \frac{\sqrt{x^3}}{2} - \frac{1}{\sqrt[4]{x^3}}\,dx = \frac{1}{2}\int \sqrt{x^3}\,dx - \int x^{-\frac{3}{4}}\,dx = \frac{1}{2}\cdot\frac{2}{5}x^{\frac{5}{2}} - \left(4x^{\frac{1}{4}}\right)$

$= \frac{1}{5}x^{\frac{5}{2}} - 4x^{1/4} + C$

2. e) $\int \frac{3x^4}{\sqrt{2x^5-4}}\,dx = \int \frac{3\,du}{10\sqrt{u}} = \frac{3}{10}\int u^{-\frac{1}{2}}\,du = \frac{2\sqrt{x^5}}{5} - 4\sqrt[4]{x} + C$

$u = 2x^5 - 4$

$du = 10x^4\,dx$

$\frac{du}{10} = x^4\,dx$

$= \frac{3}{10}\cdot 2u^{1/2} = \frac{3}{5}u^{1/2} = \frac{3}{5}\sqrt{2x^5-4} + C$

f) $\int e^{\operatorname{sen}(x)}\cos(x)\,dx = \int e^u\,du = e^u = e^{\operatorname{sen}(x)} + C$

$u = \operatorname{sen}(x)$

$du = \cos(x)\,dx$

d) $\int \frac{e^{\frac{1}{x}}}{x^2}\,dx = \int \frac{e^u}{-du^{-1}} = -\int e^u\,du = -e^u + C$

$= -e^{1/x} + C$

$u = 1/x$

$-du = \frac{1}{x^2}$

3. LIATE

a) $\int x \cdot \operatorname{sen}(x)\,dx$

c) $\int (2x-1)\cdot\cos(x)\,dx = \operatorname{sin}(x)(2x-1) - \int 2\operatorname{sin}(x)dx$

$= \operatorname{sin}(x)(2x-1) + 2\cos(x) + C$

derivar $\begin{cases} u = 2x-1 & dV = \cos(x)\,dx \\ du = 2dx & V = \sin(x) \end{cases}$ integrar

$\int e^x \cdot \operatorname{Sen}(x)\,dx =$

Cíclica

4. d) $\int \cos^2(x)\operatorname{sin}^7(x)\,dx = \int \cos^2(x)(\operatorname{Sen}^2(x))^3 \operatorname{sin}(x)\,dx$

$= \int \cos^2(x)(1-\cos^2(x))^3 \operatorname{sin}(x)\,dx = -\int u^2(1-u^2)^3 du$

$u = \cos(x)$

$-du = \operatorname{sin}(x)dx$

$= -\int u^2(1^3 - 3\cdot 1^2\cdot u^2 + 3\cdot 1\cdot u^4 - u^6)du = -\int u^2(1-3u^2+3u^4-u^6)du$

$= -\int (u^2 - 3u^4 + 3u^6 - u^8)du = -\left[\int u^2du - 3\int u^4du + 3\int u^6du - \int u^8du\right]$

$= -\left(\frac{u^3}{3} - \frac{3u^5}{5} + \frac{3u^7}{7} - \frac{u^9}{9}\right) = -\left(\frac{\cos^3(x)}{3} - \frac{3\cos^5(x)}{5} + \frac{3\cos^7(x)}{7} - \frac{\cos^9(x)}{9}\right)$

$+ C$

4. b) $\int \cos^2(x)\sin^2(x)\,dx =$

$= \frac{1}{4}\left(\int 1\,dx - \int \cos^2(2x)\,dx\right) \cdots$

a) $\int 3\cot(x)\csc^5(x)\,dx = 3\int \cot(x)\csc^4(x)\csc(x)\,dx$

$= 3\int \cot(x)(\csc^2(x))^2\csc(x)\,dx = -3\int u^4\,du = -3\frac{u^5}{5} = \frac{-3\csc^5(x)}{5}$

$u = \csc(x)$

$-du = \cot(x)\csc(x)$

5. $x^2 - 10x + 34 = -34$

$x^2 - 10x + 25 = -9$

$(x+5)^2 + 9 =$

$\int \frac{3}{x^2 - 10x + 34}\,dx$

$256 : 6 = 42$

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b. $\int \frac{\sqrt{x^2-16}}{x^4}\,dx = \int \frac{\sqrt{16\sec^2(\theta) - 16}}{256\sec^4(\theta)} 4\sec(\theta)\tan(\theta)\,d\theta = \frac{4}{256}\int \frac{\sqrt{16(\sec^2(\theta)-1}}{\sec^4(\theta)}\,dx$

$x = 4\sec(\theta)$

$dx = 4\sec\theta\tan\theta\,d\theta$

$= \frac{8}{256}\int \frac{\sqrt{\sec^2(\theta)-1}}{\sec^4(\theta)}\,dx = \frac{1}{32}\int \frac{\tan^2(\theta)\sec(\theta)}{\sec^4(\theta)} = \frac{1}{16}\int \sin^2(\theta)\cos(\theta)\,dx$

6. $\int x\sqrt{x^2+9}\,dx = \int 3\tan(\theta)\sqrt{9\tan^2(\theta)+9}\;3\sec^2(\theta)\,d\theta$

$x = 3\tan(\theta)$

$dx = 3\sec^2(\theta)\,d\theta$

$= 27\int \tan(\theta)\sec^3(\theta)\,d\theta = 27\int \frac{\sin(\theta)}{\cos(\theta)}\cdot\frac{1}{\cos^3(\theta)}\,d\theta$

$= 27\int \frac{\sin(\theta)}{\cos^4(\theta)}\,d\theta = -27\int u^{-4}\,du = +\frac{27}{3}u^{-3} = 9u^{-3}$

$u = \cos(\theta)$

$-du = \sin(\theta)$

$= 9\cos^{-3}(\theta)+C = 9\cdot\frac{1}{\cos^3(\theta)}+C$

$= \frac{9}{\left(\frac{3}{\sqrt{x^2+9}}\right)^3}+C$

(op) / h

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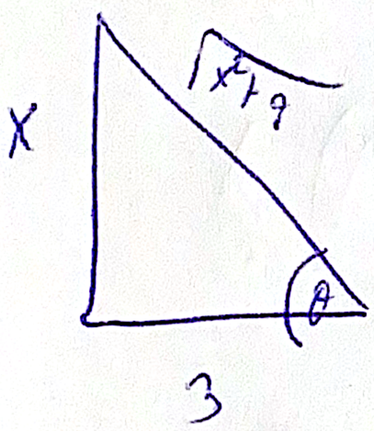

5. $\int \frac{\cos(x)}{1+\sin^2(x)} = \int \frac{1}{1+u^2}\,du = \arctan(u)+C$

$= \arctan(\sin(x))+C$

$u = \sin(x)$

$du = \cos(x)$

d) $\int \frac{(5x-3)}{x(x-2)^2}\,dx = \frac{A}{x} + \frac{B}{x-2} + \frac{C}{(x-2)^2}$

$5x-3 = A(x-2)^2 + Bx(x-2) + C(x)$ / si $x=2$

$7 = C2 \rightarrow C = 7/2$

/ si $x=0$

$4 = 4C + 2$

$-3 = 4A + 0$

$\frac{-3}{4} = A$

$2 = \frac{-3}{4}(1) + -B + \frac{7}{2}$

$\frac{11}{4} + -2 = B$

$\frac{11}{4} - 2 = B$

$\frac{11-8}{4} = B \Rightarrow B = \frac{3}{4}$

$\int \frac{(5x-3)}{x(x-2)^2}\,dx = \int \left(\frac{-3/4}{x} + \frac{3}{4} + \frac{7/2}{(x-2)^2} \right) dx$

$u = x-2 \Rightarrow du = dx$

$\int u^{-2}\,du = -u^{-1}$

$\frac{3}{4}\int \frac{1}{x}\,dx + \frac{3}{4}\int \frac{1}{x-2}\,dx + \frac{7}{2}\int \frac{1}{(x-2)^2}\,dx$

$= -\frac{3}{4}\ln|x| + \frac{3}{4}\ln|x-2| + \frac{7}{2}\cdot -(x-2)^{-1} + C$

e) $\int \tan(x)\sec^2(x)\,dx = \int \frac{\sin(x)}{\cos(x)\cos^2(x)}\,dx = \int \frac{\sec(x)\tan(x)}{\sec(x)\tan(x)}\,dx$

$u = \sin \quad du = \cos dx$
$u = \cos \quad du =$

$\int u\,du$

$u = \cos(x)$
$du = \sin(x)\,dx$

$$\int u^{-3}\,du = \frac{-u^{-2}}{2} + C$$

$$= -\frac{\cos^{-2}(x)}{2}$$